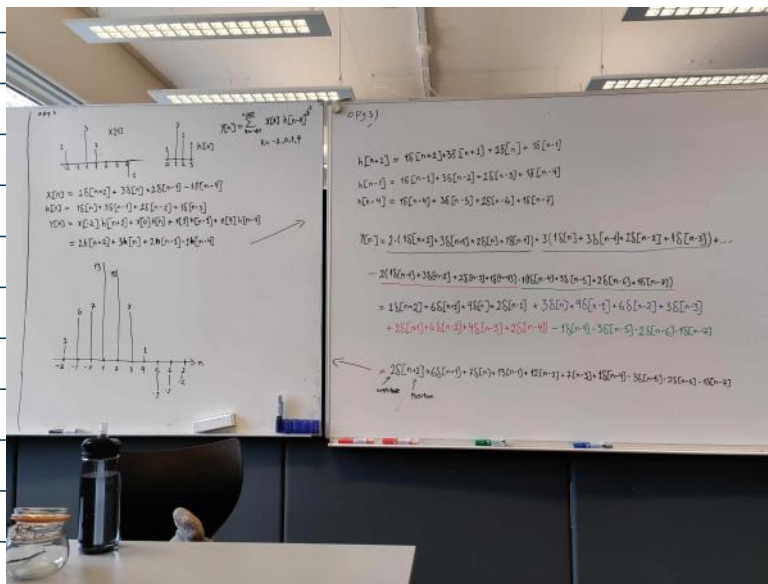


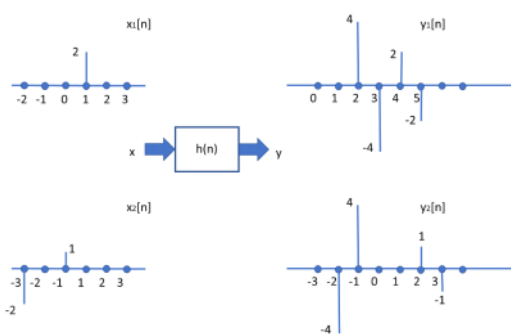
Lec 9

Friday, 30 May 2025 10.20

Opg 3)



4. For a system having response $h[n]$, the following relationship exists between two input and output signals:



Can h be a linear time invariant system? Is h causal?

opg 4)

Skal tjekke hvad $h[n]$ er. Skal arbejde baglens p. hvad vi gjorde i opg 3

2) If h represents an LTI system, then

$$y_1[n] = \sum_{k=-\infty}^{+\infty} x_1[k] h[n-k] = 4\delta[n-2] - 4\delta[n-3] + 2\delta[n-4] - 2\delta[n-5] \quad \text{Seksens for } y_1[n]$$

$$\sum_{k=-\infty}^{+\infty} x_1[k] h[n-k] = x_1[1] h[n-1] = 2h[n-1] \quad \text{Tager og finder hvad h burde shifte med (ud fra k varden)}$$

$$2h[n-1] = 4\delta[n-2] - 4\delta[n-3] + 2\delta[n-4] - 2\delta[n-5] \quad \text{Dividert amp}$$

$$h[n-1] = 2\delta[n-2] - 2\delta[n-3] + \delta[n-4] - \delta[n-5] \quad \text{Shifter med 1}$$

$$h[n] = 2\delta[n-1] - 2\delta[n-2] + \delta[n-3] - \delta[n-4] \quad \text{h[n] er fundet til dette}$$

We need to verify that - Bruger nu den fundne $h[n]$ til at verif.

$$y_2[n] = \sum_{k=-\infty}^{+\infty} x_2[k] h[n-k] = x_2[-3] h[n+3] + x_2[-1] h[n+1] =$$

$$= -4\delta[n+2] + 4\delta[n+1] + \delta[n-2] - \delta[n-3] \quad \text{lavet nu convolution } \sum_{k=-\infty}^{+\infty} x_2[k] h[n-k].$$

Hvis dette resultat er $y_2[n]$ så ved vi $h[n]$ er korrekt.

Seksens af $y_2[n]$

$$\text{For } -3: x_2[-3] h[n+3] = -2h[n+3] = -2(2\delta[n+2] - 2\delta[n+1] + \delta[n-2] - \delta[n-3])$$

$$= -4\delta[n+2] + 4\delta[n+1] - 2\delta[n-2] + 2\delta[n-3]$$

Summer disse sammen

$$\text{For } -1: x_2[-1] h[n+1] = h[n+1] = 2\delta[n] - 2\delta[n-1] + \delta[n-2] - \delta[n-3]$$

$$y_2[n] = -4\delta[n+2] + 4\delta[n+1] + \delta[n-2] - \delta[n-3] \quad \checkmark$$

h can then be an LTI system!

Giver det samme som den tidligere $y_2[n]$ seksens.
 $h[n]$ må der

Since $h[n]$ does not depend of future values of n , the system is causal.